

JAMES ALEXANDER PATTERSON

Liverpool, United Kingdom

Email: james.patterson@gmail.com

Phone: +44 7911 584293

LinkedIn: linkedin.com/in/jamespatterson-ml

GitHub: github.com/jamespatterson-ml

PROFESSIONAL SUMMARY

Results-driven Machine Learning Engineer with 5 years of progressive experience building and deploying ML solutions across fintech and data analytics sectors. Proven expertise in Python, TensorFlow, and cloud platforms with a strong track record of translating business requirements into scalable machine learning models. Passionate about optimizing model performance and mentoring junior team members while staying current with emerging ML technologies and best practices.

WORK EXPERIENCE

Senior Machine Learning Engineer

DataVault Solutions Ltd

<https://www.datavault.co.uk>

Liverpool, United Kingdom

March 2022 - Present

Designed and implemented a customer churn prediction model using XGBoost that reduced customer attrition by 18 percent, generating GBP 450,000 in retained annual revenue.

Led the migration of legacy Python 2 ML pipelines to production-ready Python 3 microservices deployed on AWS Lambda, reducing inference latency by 35 percent.

Established MLOps best practices including model versioning, automated testing, and CI/CD pipelines using Jenkins and Docker, improving deployment frequency from monthly to weekly releases.

Mentored three junior machine learning engineers on feature engineering, hyperparameter tuning, and model evaluation techniques, with all team members successfully delivering independent projects.

Machine Learning Engineer

Neuron Analytics Inc

<https://www.neuronanalytics.io>

Liverpool, United Kingdom

July 2020 - February 2022

Developed and trained deep learning models using TensorFlow and Keras for time-series forecasting of financial market trends, achieving 92 percent prediction accuracy on test datasets.

Built data preprocessing pipelines using Apache Spark to handle datasets exceeding 500GB, reducing processing time from 6 hours to 45 minutes through distributed computing optimization.

Collaborated with product and data science teams to define success metrics and conducted A/B testing on model variants, selecting the optimal approach for production deployment.

Created comprehensive documentation and Jupyter notebooks explaining model architecture and hyperparameter choices, enabling knowledge transfer across the analytics department.

Junior Machine Learning Engineer

TechCore Innovations Ltd

<https://www.techcoreinnovations.net>

Manchester, United Kingdom

June 2019 - June 2020

Developed classification models using scikit-learn for customer segmentation, enabling the marketing team to create targeted campaigns that increased conversion rates by 12 percent.

Performed exploratory data analysis on structured and unstructured datasets using Pandas, NumPy, and Matplotlib to identify patterns and inform feature selection decisions.

Implemented automated data quality checks and anomaly detection systems that flagged data integrity issues in real-time, preventing 8 instances of model degradation.

Participated in code reviews and contributed to an internal Python ML library used by 15 engineers, submitting 23 pull requests with cumulative improvements in code efficiency and readability.

EDUCATION

Bachelor of Science in Computer Science

University of Manchester

Manchester, United Kingdom

2015 - 2019

Grade: 2:1 (Upper Second Class Honours)

SKILLS

Programming Languages: Python, SQL, R, Java, Scala

Machine Learning Frameworks: TensorFlow, Keras, PyTorch, scikit-learn, XGBoost

Data Processing: Apache Spark, Pandas, NumPy, PySpark

Cloud Platforms: AWS (EC2, S3, SageMaker, Lambda), Google Cloud Platform (BigQuery, Vertex AI)

MLOps and DevOps: Docker, Kubernetes, Jenkins, Git, GitHub Actions

Data Visualization: Matplotlib, Seaborn, Plotly, Tableau

Databases: PostgreSQL, MongoDB, Redis, Elasticsearch

Additional: Jupyter Notebooks, Linux, REST APIs, Agile/Scrum

PUBLIC REPOSITORIES

Repository 1: Time-Series-Forecasting-Pipeline

<https://github.com/jamespatterson-ml/time-series-forecasting-pipeline>

LSTM neural network implementation for multivariate time-series prediction with data preprocessing and model evaluation utilities.

Repository 2: Customer-Churn-Prediction-Model

<https://github.com/jamespatterson-ml/customer-churn-prediction>

End-to-end binary classification project including exploratory data analysis, feature engineering, hyperparameter optimization, and Flask API deployment.

Repository 3: ML-Pipeline-Automation-Framework

<https://github.com/jamespatterson-ml/ml-pipeline-automation>

Reusable framework for automating common ML workflows including data validation, model training, cross-validation, and performance monitoring.

Repository 4: Computer-Vision-Object-Detection

<https://github.com/jamespatterson-ml/cv-object-detection>

YOLOv5 implementation for real-time object detection with custom dataset integration and evaluation metrics.

CERTIFICATIONS

AWS Certified Machine Learning Specialist

Issued: August 2021, Expires: August 2024

Deep Learning Specialization

Coursera, Completed: December 2020

LANGUAGES

English: Native

Spanish: Intermediate